

Name: ----- ID No ----- Campus/Branch-----

General Instruction

- This model exit exam contains a total of 156 multiple choice questions. Make sure that the exam contains a total of twenty six pages.
- When you seat in this exam, you should switch off your cell phone and using any electronic equipment is strictly forbidden.
- A student who engages in cheating and tries to cheat by any means automatically results zero in his/her score.
- For your answer use only upper cases and provide the correct answer on the provided space only.
- Do not detach any paper of this exam until you told to do so. Using additional white paper is strictly forbidden.
- Time allowed 3:00 Hrs.

Do not turn this page until you told to do so!!!

Good Luck!!!

Instruction: Choose the best answer from the given alternatives

1. At which stage of staffing process should whether job description and job specification are matched be checked?
A. Induction and orientation C. Selection

- B. Placement
D. Recruitment
2. A formal pattern of interaction and coordination designed by management to link the task of individuals and groups to achieve organizational goal and to show reporting relationship refers to us_____?
- A. Organizational structure
C. Organization
B. Staffing
D. Leadership
3. In addition to its basic functions, an organization may have supportive and/or advisory units as a staff authority. Which one of the following is a staff authority?
- A. Production/operation
C. Accounting and finance
B. Marketing
D. Research and development
4. Which one of the following is FALSE about management?
- A. Management brings together factors of production to generate output
B. Management is a bottleneck to achieve operational efficiency and effectiveness
C. The economic disparity among countries of the world lies basically in their proper understanding and application of management to exploit their natural and human resources for their betterment
D. Management involves properly practicing the basic managerial functions
5. Human resource surplus can be addressed through the following mechanisms EXCEPT?
- A. Hiring freeze
C. Hiring
B. Layoff
D. Forced retirement
6. Leadership is a very important aspect of an organization. How does effective leadership contribute to organizational success?
- A. It disintegrates group effort
B. It results in unfair resource allocation
C. It helps to direct organised efforts towards a common goal
D. It reduces organizational efficiency

7. Mr. 'X' is a manager of 'ABC' company. He usually tours visitors of the company, signs business deals and takes some important customers to lunch. Which managerial role is Mr. 'X' playing?
- A. Figurehead
B. Decisional
C. Spokes person
D. Liaison
8. Which one of the following is a type of plan that can be used again and again over a long period of time?
- A. Program
B. Project
C. Budget
D. Policy
9. Decision making under certainty is characterised by the following EXCEPT?
- A. Complete information about the problem
B. Probability of occurrence of the various states of nature can be developed
C. Level of ambiguity is high and probability of making a good decision is very low
D. Complete information about the available choices
10. Proactive and reactive control takes place at the _____ and _____ of a process respectively?
- A. Beginning, end
B. End, beginning
C. Beginning, middle
D. Middle, end
11. Which one of the following is not the correct description of modern marketing?
- A. Marketing is the creation of value for customers.
B. Marketing is managing profitable customer relationships.
C. Selling and advertising are synonymous with marketing.
D. Marketing involves satisfying customers' needs.
12. Which one of the following philosophy holds that consumers favor those products that offer the most quality, performance, or innovative features?
- A. Product philosophy
B. Production Philosophy
C. Selling Philosophy
D. Marketing philosophy

13. Mr X has been directed to study the demographic, economic, political, and cultural forces that affect an organization. In this instance, which of the following one Mr X has been directed to study?
- A. Macro environment
 - B. Micro environment
 - C. Internal environment
 - D. Marketing mix
14. Companies can research many aspects of buying decisions. Which one is the most difficult to identify?
- A. What consumers buy
 - B. Why they buy
 - C. When they buy
 - D. Where they buy
15. Under which type of buying decision the consumer has some beliefs about cookies, chooses a brand of cookies without much evaluation, and evaluates the product during consumption?
- A. Complex buying behavior
 - B. Dissonance – reducing buying behavior
 - C. Habitual buying behavior
 - D. Variety seeking buying behavior
16. From the following promotion mix elements one is not true?
- A. Advertising is any Paid form of non-personal presentation and promotion of ideas, goods, or services by an identified sponsor
 - B. Direct Marketing is use of mail, telephone, fax, e-mail, and other non-personal contact tools to communicate directly with specific customers and prospects.
 - C. Personal selling is a variety of short-term incentives to encourage trial or purchase of a product or service.
 - D. Publicity: A variety of programs designed to promote and/or protect a company's image or its individual products.
17. One of the following is true about method determine the selling price?
- A. Cost plus pricing means setting the price of one unit of a product equal to the total cost of the unit plus the desired profit on the unit.
 - B. Dumping is a form of price discrimination, is the practice of charging different price for the same product in similar markets.
 - C. Market penetration Pricing Setting a relatively high initial price for a new product is referred to as market skimming pricing.

- D. Market Skimming pricing, a relatively low initial price is established for a new product.
18. Of the method of payment one is used goods are shipped but ownership is retained by the seller?
- A. Consignment
 - B. Cash in Advance
 - C. Letter of Credit
 - D. Bill of exchange
19. Which one of the following is not indirect channel of distribution?
- A. Foreign distributor
 - B. Foreign Retailer
 - C. State-control Trading Company
 - D. Resident Buyer
20. In any domestic market the international market requires a marketer to make at least three channel decisions except?
- A. Length of channels of distribution
 - B. Width of channels of distribution
 - C. Market location of channels distribution
 - D. Number of channels of distribution
21. One is a more narrowly defined group, typically a small market whose needs are not being well served. which one of the following is?
- A. Niche market
 - B. Segment market
 - C. Mass market
 - D. Local market
22. Which one is a good that are normally consumed in one or a few uses, dealers make them available in many locations, charge only a small markup, and advertise heavily to induce trial and build preference?
- A. Durable
 - B. Nondurable goods
 - C. Consumer goods
 - D. Industrial goods
23. Which one of the following is any group that has an actual or potential interest in, or impact on, an organization's ability to achieve its objectives?
- A. Team
 - B. Competitor
 - C. Market
 - D. Public

24. Among the elements the company has to consider in product pricing the most fundamental one is?
- A. Profit margin of retailers
 - B. Profit margin of wholesalers
 - C. Cost of product
 - D. Government tax
25. What ends all channels?
- A. Consumer
 - B. Retailer
 - C. Wholesaler
 - D. Manufacture
26. Which one of the following is **false** about domestic market?
- A. Relatively stable business environment
 - B. Market is much more homogeneous
 - C. Minimum government interference
 - D. Multiple business environments
27. One of the following is form of entering in to international market by carrying on export activities on behalf of several producers and partially under their administration?
- A. Export – management company
 - B. Domestic based export agent
 - C. Cooperative organization:
 - D. Domestic – based export merchant
28. Firms must undertake an environmental analysis before making a decision for International marketing EXCEPT?
- A. PEST Analysis
 - B. Competitive Intelligence
 - C. SWOT Analysis
 - D. Five Forces
29. Which one of the following product development process one is next to business Analysis
- A. Prototype development
 - B. Screening ideas
 - C. Market tests
 - D. Commercialization

30. All of the major product mix strategies that the producers and middlemen need carefully planned strategies EXCEPT?
- A. Positioning
 - B. Alteration
 - C. Expansion
 - D. Advertising
31. -----it is a widely used recruitment method in order to reach relatively large number of potential applicants through daily newspapers, trade and professional newspapers, and occasionally on radio, TV and billboards.
- A. Job Advertisement
 - B. Job Posting
 - C. Job auditing
 - D. Job creation
32. -----is an attempt to forecast how many and what kind of employees will be required in the future and to what extent this demand is likely to be met.
- A. Job analysis
 - B. Management development
 - C. Human resources inventory
 - D. Human resource planning
33. It is a unique capability in the organization that creates high value and that differentiates the organization from its competitors is:
- A. Core Competency
 - B. Opportunity
 - C. Objective
 - D. Strategy
34. One is not the solution to overcome when human resource surplus exists in an organization?
- A. Retention
 - B. Voluntary departure
 - C. Early retirement
 - D. Hiring freeze
35. It is a horizontal expansion of tasks, or duties assigned to a given job is____
- A. Job Rotation
 - B. Job Enlargement
 - C. Job enrichment
 - D. Job specification
36. An HR function which practiced to increasing the technical skill of employees useful for the current job is called_____.
- A. Employee training
 - B. Employee staffing
 - C. management development
 - D. Employee development
37. Among the factors influence compensation amount determination, which force not contribute the organization to raise the amount of payment for its employees?
- A. When an organization running at loss
 - B. When the working environment is hazardous
 - C. When there is complex and challenge types of tasks
 - D. When there is strong and influential union

38. It involves assisting employees in defining their occupational interests, and preparing for job changes through human resource development is _____.
A. Manpower planning
B. Training
C. Development
D. Career Planning
39. Which of the following not main objectives of collective bargaining?
A. To maintain cordial relations between the employer and the employees.
B. To ensure the participation of trade unions in industry.
C. To avoid the need for Government intervention as collective bargaining is a voluntary process.
D. To demote industrial democracy
40. Which of the following is not the barrier of selection associated with validity of measurement instrument?
A. When the proposed test can measure what it supposed to measure
B. When the proposed test produce consistent result when it repeated in similar situation
C. When the proposed test cannot measure what it supposed to measure
D. When the proposed test measured the need of skill or knowledge required to operate job
41. Organizational Commitment is
A. The degree, to which a person identifies his job, actively participates in it, and considers his performance important to self-worth.
B. The evaluative statements either favorable or unfavorable concerning objects, people, or events.
C. The degree to which an employee identifies organization and its goals particularly and wishes to maintain membership in the organization.
D. The sum total of ways in which an individual reacts and interacts with others.
42. Which of the following is not true about matrix structure of organizations?
A. The organization places the employees based on the function and the product.
B. Organizations divide the functional areas of the organization to divisions.
C. The matrix structure gives the best of the both functional and divisional structures.
D. In matrix structure, the company uses teams to complete tasks.

43. The process of receiving, selecting, organizing, interpreting, checking, and reacting to sensory stimuli or data is
- A. Perception
 - B. Attitude
 - C. Personality
 - D. Learning
44. Organizational behavior analyzes three levels of an organization-which are
- A. Country, employer, society
 - B. Society, country, nation
 - C. Employee, employer, management
 - D. Individuals, groups, organization
45. The following are the importance of motivation except
- A. To attain high performance level
 - B. To maximize employee's turnover
 - C. To increase work efficiency
 - D. To increase willingness to work
46. Assume that the production department of a manufacturing company may have a dispute with quality control because new quality procedures reduce production efficiency. Such type of conflict is categorized under:
- A. Inter-personal conflict
 - B. Inter-group conflict
 - C. Intra-personal conflict
 - D. Inter-organizational conflict
47. Which one of the following statement is false?
- A. It is easier for management to deal with resistance when it is overt and immediate
 - B. Win lose orientation is a systematic attempt to maximize the goals of one party
 - C. Restraining forces are forces that push organizations toward a new state of affairs
 - D. Reengineering is not making marginal improvements to the business
48. Which one of the leadership style is likely be can apply in organizations with highly skilled and well-trained professional?
- A. Autocratic leadership
 - B. Democratic leadership
 - C. Free rein leadership
 - D. Situational leadership

49. Which of the contingency/situational leadership theory claim that the performance of groups is dependent on the interaction between leadership style and situational favourableness?
- A. Fiedler's contingency model (Fred E. Fiedler)
 - B. Leadership continuum theory (Robert Tannenbaum and Warren Schemdit)
 - C. Situational leadership model (Paul Hersey and Ken Blanchard)
 - D. Transformational leadership (James Burns and Bernard M. Bass)
50. Among the planned organizational change; discovery, dreaming, designing, and delivering are mentioned under:
- A. Kurt Lewin change model
 - B. Action research approach
 - C. Lessier's change model
 - D. Appreciate inquiry approach
51. Among the following human resource management function which one concern about familiarizing or socializing the new employees to the overall organization
- A. Placement
 - B. Orientation
 - C. Induction
 - D. Training and development
52. Of the following one is not the importance of effective performance appraisal
- A. To determine who shall receive merit increase
 - B. To determine training need
 - C. To discontent employees of the organization
 - D. To provide adequate feedback to each individuals for his/her performance
53. Which of the following NOT indirect compensation management practices?
- A. Insurance
 - B. House Rent Allowance
 - C. Holiday Homes
 - D. Flexible Timing
54. Which one **not** is the objective of compensation management practice in organization?
- A. To establish a fair and equitable remuneration practice in organization
 - B. To attract competent personnel from other organization
 - C. To retain the inept present employees
 - D. To improve public image of the company
55. Which of the **not** internal factors affecting human resource management function?
- A. Labour market conditions
 - B. Organization Policies
 - C. Management Style of Upper Managers
 - D. Corporate Culture

56. If you admire someone to the point of modeling your behavior and attitudes after him or her, that person possesses which type of power?
- A. Reward Power
 - B. Legitimate Power
 - C. Expert Power
 - D. Referent Power
57. The pattern of shared values, beliefs, and assumptions considered for the appropriate way to think and act within an organization is
- A. Organizational culture
 - B. Organizational conflict
 - C. Organizational behavior
 - D. Organizational politics
58. Job dissatisfaction is the simplest and most obvious psychological effect of stress. This consequence of stress in organizations is
- A. Physiological symptoms
 - B. Behavioral symptoms
 - C. Psychological symptoms
 - D. Employee empowerment
59. Which of the following is true about compromising conflict management technique?
- A. It is a negotiation style where one party is willing to oblige or adapt to meet the needs of the other party.
 - B. It occurs when one party negotiates to maximize its results at the expense of the other party's needs.
 - C. It is the settlement of differences through concessions of one or both parties.
 - D. It occurs when people cooperate to produce a solution satisfactory to both.
60. Which type of group development stage is characterized by close relationship and cohesiveness among group members?
- A. Forming
 - B. Norming
 - C. Storming
 - D. Performing
61. If Dashen brewery S.C. started producing malt products in order to satisfy its malt need of the market, which strategy does the company applied?
- A. Horizontal integration
 - B. Backward integration
 - C. Conglomerate diversification
 - D. Concentric diversification

62. At which stage of new product development management prepares sales, cost and profit projections to determine whether they satisfy the company's objective or not?
- A. Commercialization
B. Business Analysis
C. Market Testing
D. Idea Screening
63. A form of communication that seeks to change the perceptions of customers and other publics about a company and its products is
- A. Sales promotion
B. Public relation
C. Personal selling
D. Advertising
64. Among the following pricing methods one is psychological pricing method?
- A. Cost-plus pricing
B. Mark-up pricing
C. Competition Oriented Pricing
D. Odd-even pricing
65. A component of marketing research which is conducted to know how favourably consumers view the brand is said to be_____.
- A. Buyer decision processes research
B. Brand equity research
C. Market penetration
D. Market Share
66. One of the following has a problem of double taxation.
- A. Partnership
B. Corporation
C. Sole proprietorship
D. Cooperatives
67. Which of the following is not true about knowledge management?
- A. Knowledge is just like light and intangible asset of a given organization
B. Knowledge informs decisions and actions.
C. Without knowledge organization can survive.
D. Without data and information there is no knowledge.
68. Identify the correct order from the smallest to the largest.
- A. Byte □□Bite □□Word □□Field □□File □□Database
B. Bite □□Byte □□Word □□Field □□File □□Database
C. Bite □□Byte □□Word □□File □□Field □□Database
D. Byte □□Bite □□Word □□File □□Field □□Database

69. How many four wheel green- cars found in Addiss Ababa are blue? which indicates --
- A. Backup
 - B. Query ability
 - C. Data sequencing
 - D. Data routing
70. Which of the following is **not** the benefits of decision support system
- A. Replace human creativity and experience.
 - B. Augment decision maker's knowledge management abilities.
 - C. Allow decision maker to solve larger and more complex problems.
 - D. Make decision making faster and more reliable.
71. Which one of the following is not the characteristic of reengineering?
- A. It is about rethinking work from the ground up in order to eliminate work that is not necessary
 - B. It focuses on how an organization is structured around an organizational chart or business unit.
 - C. It simply determines what work needs to be done and finds the best way to accomplish it.
 - D. It focus is the customer not automation or computerization.
72. It is the type of leadership based on the principle of "you do this work for me and I will give this reward to you."
- A. Transactional leadership
 - B. Transformational leadership
 - C. Servant leadership
 - D. Autocratic leadership
73. Which of the conflict management style involves making unilateral concessions and unconditional promises, as well as offering help with no expectation of reciprocal help?
- A. Avoiding
 - B. Forcing
 - C. Yielding
 - D. Compromising
74. The least risky strategy to stimulate conflict in the organization is:
- A. Communication
 - B. Stimulating competition
 - C. Changing organizational structure
 - D. Bringing an outside person
75. Which one of the following is not type of revolutionary change?

- A. Reengineering
- B. Restructuring
- C. Total quality management
- D. Quantum innovation

76. One of the following is not included in the process of entrepreneurship?

- A. Perceiving an opportunity.
- B. Commercializing perceived opportunity
- C. Fear of risk or failure.
- D. Creativity

77. Assume Kombolcha Textile Factory buys cotton farm and begins to supply inputs to its own, the strategy followed by the factory is said to be _____.

- A. Forward integration
- B. Backward integration
- C. Horizontal integration
- D. Concentric diversification

78. Which of the following is not among the environmental factors which hinder entrepreneurial growth?

- A. Sudden changes in government policy.
- B. Sudden political upsurge.
- C. Excessive bureaucracy and corruption.
- D. Political stability or optimistic government attitude towards industry.

79. All of the followings are sources of equity capital for a firm EXCEPT:

- A. Personal Savings
- B. Friends and Relatives
- C. Bond
- D. Stock

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- A. Partnership
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86. Which of the following is not true about knowledge management?

- A. Knowledge is just like light and intangible asset of a given organization
- B. Knowledge informs decisions and actions.
- C. Without knowledge organization can survive.
- D. Without data and information there is no knowledge.

87. A sound system of automated and manual internal controls must exist throughout all information systems processing activities. Which feature does it indicate?

A. Completeness B. Accuracy C. Consistency D. Timeliness E. Relevance.

88. _____ Cables carry more information, suffer less interference, and require fewer signal repeaters over long distances than wires.

A. Wire Pairs B. Coaxial Cables
C. Fiber Optics D. Microwaves

89. _____ systems provide a powerful technology for tracking the movement of goods throughout the supply chain.

A. Radio frequency identification (RFID)
B. Broadband
C. Router
D. CPU

93. _____ is used when the problem is complex and the information needed to make the best decision is difficult to obtain and use.

A. MIS
B. DSS
C. EIS
D. TPS

94. Which one of the following is the correct order?
- A. Information , Data , Wisdom , Knowledge
 - B. Data, Information, Knowledge , Wisdom
 - C. Data, Information, Wisdom, Knowledge
 - D. Information, Data, Knowledge, Wisdom
95. _____work with users to define the requirements for developing new systems and improving existing ones.
- A. System analysts
 - B. Database administrators
 - C. Network Specialists
 - D. Programmers
96. Organizations pursue horizontal integration because :
- A. They compete in a growing industry
 - B. Increased economies of scale provide them major competitive advantages
 - C. Competitors are faltering due to a lack of managerial expertise
 - D. All are the reason why they go for the stated strategy
97. The intensity of rivalry among established companies within an industry is largely a function of :
- A. Industry competitive structure;
 - B. Demand conditions;
 - C. Cost conditions; and
 - D. The height of exit barriers in the industry.
 - E. All
98. Bargaining Power of consumer is high when,
- A. buyers can inexpensively switch
 - B. buyers are particularly important
 - C. sellers are struggling in the face of falling consumer demand
 - D. All
99. _____ is a statement that answers the question, “What do we want to become?”
- A. Mission
 - B. Goals
 - C. Vision
 - D. Tactics
100. _____ is a statement that addresses the basic question that faces all strategists, “What is our business?”
- A. Mission
 - B. Goals
 - C. Vision
 - D. Tactics

101. Strategy formulation involves all of the following, EXCEPT:
- A. Identifying a vision and mission statements
 - B. Situation analysis
 - C. Generating alternative strategies
 - D. Allocating resources
102. A successful external environmental audit will result in :
- A. Environmental opportunities
 - B. Environmental threats
 - C. Environmental strengths
 - D. A and B
103. Defensive strategies include the following, EXCEPT:
- A. Market penetration strategy
 - B. Retrenchment strategy
 - C. Divestiture strategy
 - D. All
104. Forward integration involves taking control of _____.
- A. Suppliers of raw materials
 - B. Distributors of products
 - C. Both suppliers and distributors
 - D. None of the above
105. Backward integration is suitable
- A. When an organization's present suppliers are especially expensive or unreliable
 - B. When the number of suppliers is small and the number of competitors is large
 - C. When an organization's present distributors are especially expensive
 - D. A and B
106. Which one of the following statements best describes a project?
- A. A project is a set of tools and techniques often used when delivering organizational change.
 - B. A project is the sum of activities needed to remove uncertainty from a unique piece of work.
 - C. A unique transient endeavor undertaken to achieve a desired outcome.
 - D. A project is a method of planning work.
107. Which of the following ensure projects are soundly designed and appropriately engineered?
- A. Institutional analysis
 - B. Economic analysis
 - C. Technical analysis
 - D. Financial analysis.

108. BGI designed a project to acquire Meta brewery, what is the source of project idea for BGI?
- A. Government
 - B. Constraint
 - C. Resource
 - D. Linkage
109. According to Baum model of project cycle what is the first stage:
- A. Investment
 - B. Evaluation
 - C. Preparation
 - D. Identification
110. Which one of the following cannot be the rationale for social cost benefit analysis?
- A. Market Imperfections
 - B. Externalities
 - C. Concern for saving
 - D. None of the above
111. Which one of the following is FALSE about market and demand analysis?
- A. Estimate the potential size of the market for the proposed product.
 - B. Give insights about the market share to be captured.
 - C. Identify aggregate demand for the product.
 - D. It has effect on the technical analysis.
112. Network Diagram is best applicable when project tasks:
- A. Are well defined and have clear beginning and end point
 - B. Are inter-related
 - C. Are very large and complex
 - D. All of the above
113. All are the need for monitoring and evaluation, EXCEPT
- A. To know to what extent project is meeting its objectives.
 - B. To learn lessons from project experience.
 - C. Provide a clear basis for decision-making.
 - D. None of the above
114. Project evaluation technique that sum up the achievements, impact and lessons learned:
- A. Midterm evaluation
 - B. Outcome evaluation
 - C. Periodic evaluation
 - D. Formative evaluation
115. Benefit Cost Ratio (BCR)>1 indicates which of the following;

- A. Benefits exceed costs
 - B. Positive NPV
 - C. Benefits and costs are in balance
 - D. Costs exceed benefit
116. All are prove the scientific methods of research, except?
- A. It results into non probabilistic predictions
 - B. It relies on empirical evidence;
 - C. It utilizes relevant theories, concepts and other relevant related issues;
 - D. It is committed to only objective considerations
117. One of the following is not true about sampling design?
- A. Sample design must result in a truly representative sample.
 - B. Sample design must be results in a small sampling error.
 - C. Sample design must be viable in the context of funds available for the research study.
 - D. Sample design must be employ systematic bias in the research process
118. All are the functions of report writing, except
- A. It serves as a means for judging the quality of the completed research project.
 - B. It provides unsystematic knowledge on problems and issues analyzed
 - C. It is a means for evaluating the researcher's ability and competence to do research.
 - D. It provides factual base for formulating policies and strategies relating to the subject-matter studied.
119. One of the following is not true?
- A. If one variable depends upon or is a consequence of the other variable, it is termed as a dependent variable,
 - B. The variable that is antecedent to the dependent variable is termed as an independent variable.
 - C. The research problem is a predictive statement that relates an independent variable to a dependent variable
 - D. Confounded relationship, when the dependent variable is not free from the influence of extraneous variable(s), the relationship between them is confounded by an extraneous variable(s).

120. Which one is not true about research?
- It is the search of knowledge
 - It is scientific and systematic search for pertinent information on a specific topic.
 - It is systematized effort to gain new knowledge
 - It is the process of organizing and analyzing the data which helps to make decisions in favor of personal interests
121. All of the following are measures the variability of data, except?
- Range
 - Standard deviation
 - Mean
 - Variance
122. The religions of a person, gender of an individuals, name of an items and colour of an objects are an examples of:
- Qualitative Variables
 - Quantitative Variables
 - Continuous Variables
 - Discrete Variables
123. Suppose the total population of a small town in different categories is given as; Men = 2500, Women = 2000, Girls = 4000 and Boys = 1500. What degree that represents the girls?
- 124^0
 - 144^0
 - 148^0
 - 168^0
124. Among the following sigma notation, which one is represent the sum of: $3 + 6 + 9 + 12 + 15 + 18 + \dots + 36$
- $\sum_{i=1}^{12} 3K$
 - $\sum_{i=1}^{12} K$
 - $\sum_{i=3}^{36} 3K$
 - $\sum_{i=3}^{36} K$
125. Given the list of five pulse rates is; 70, 64, 80, 74, and 92. What is the median for this list of pulse rates?
- 74
 - 76
 - 77
 - 80
126. If the mean of six values is 15 and five of the values are 14, 16, 15, 17 and 13, what is the sixth value?
- 10
 - 13
 - 15
 - 17
127. A measure of central location provided by the value in the middle when the data are arranged in ascending order.
- Range
 - Mode
 - Mean
 - Median

128. In a class there are 70 male and 30 female students. If males averaged 72 in an examination and females averaged 60, find the mean for the entire class.
- A. 68.40 B. 64.80 C. 74.80 D. 86.40
129. Among the following statements which one is the wrong statement concerning the mode of the data set?
- A. It is not affected by extreme observations.
 B. Easy to calculate and simple to understand.
 C. It is not based on all the observations
 D. It may be different values from the set of data
130. Assume there are six different objects. What is the number of permutations of six different objects taken all together?
- A. 420 B. 520 C. 620 D. 720
131. Suppose that we have two events, A and B, with $P(A) = 0.60$, $P(B) = 0.50$, and $P(A \cap B) = 0.40$. What is the value of $P(A \cup B)$?
- A. 0.60 B. 0.70 C. 0.80 D. 0.90
132. A researcher uses systematic sampling from a company that has 480 employees listed in alphabetical order. If the desired sample size is 40 and the first name to be selected is number 10. Which of the following number will not be selected?
- A. 22 C. 182
 B. 42 D. 48
133. The probability of selecting an item in probability sampling, from the population is known and is:
- A. Equal to one C. Non zero
 B. Equal to zero D. None of the above

134. Which of the following is not a type of non-probability sampling?
- A. Quota sampling
 - B. Convenience sampling
 - C. Snowball sampling
 - D. Stratified random sampling
135. Increasing the sample size has the following effect upon the sampling error?
- A. It increases the sampling error
 - B. It reduces the sampling error
 - C. It has no effect on the sampling error
 - D. All of the above
136. A sampling frame is:
- A. A summary of the various stages involved in designing a survey
 - B. An outline view of all the main clusters of units in a sample
 - C. A list of all the units in the population from which a sample will be selected
 - D. A wooden frame used to display tables of random numbers
137. Find the 90% confidence interval estimate of μ if the sample mean $\bar{x} = 72$, the sample size is $n = 64$, and $\sigma = 12$.
- A. 71.21047 to 72.78953
 - B. 70.20561 to 73.79439
 - C. 69.53272 to 74.46728
 - D. 68.71029 to 75.28971
138. Suppose we draw a simple random sample of 500 from a large population and find that the sample proportion is 0.55. What is the 95% confidence interval estimate of the population proportion p ?
- A. 0.4937043 to 0.6062957
 - B. 0.5063936 to 0.5936064
 - C. 0.5174977 to 0.5825023
 - D. 0.4524930 to 0.6475070
139. Which of the following terms is most closely associated with the “level of significance”?
- A. One-tailed test of significance
 - B. Two-tailed test of significance
 - C. Type I Error
 - D. Type II Error

140. Range or set of values with in which population parameter is expected to occur is
- A. Central limit theorem
B. Confidence interval
C. Hypothesis testing
D. Population proportion
141. When the null hypothesis is equal to some specified value then, we use
- A. Left tailed test
B. Right tailed test
C. Two tailed test
D. One tailed test
142. A procedure based on sample evidence and probability theory to determine whether the hypothesis is a reasonable statement is
- A. Hypothesis testing
B. Estimation
C. Probability distribution
D. Normal distribution
143. Suppose a company wants to test whether a type of training is independent of the employee performance at a 5% level of significance. There for, the following table reveals that the number of employees who has taken a training and their performance.

Type of training	Employee performance			Total
	High	Medium	Low	
On the job	5	115	30	150
Off the job	20	35	45	100
Total	25	150	75	250

What is the value of the test statistic and the decision?

- A. 46.53 and reject
B. 250 and reject
C. 2.33 and accept
D. 150 and accept
144. A sport channel has 68% of spectators. What is the probability of taking a random sample of 32 people from who watch the TV channels that 0.7 or more watches the sport channel?
- A. 0.492
B. 0.008
C. 2.42
D. 0.52

145. A contractor assumes that construction workers are idle for 30 minutes or less per day. A random sample of 25 construction workers was taken and the mean idle time was found to be 50 minutes per day with a sample standard deviation of 35 minutes. Assume that the population is approximately normally distributed; use a 5% level of significance to test the contractor's assumption.
- A. Reject H_0 , 2.85 greater than 2.064
 - B. Reject H_0 , 1.96 less than 2.064
 - C. Accept H_0 , 2.85 greater than 2.064
 - D. Accept H_0 , 2.064 greater than 1.645
146. Which of the following is correct decision rule in case of ANOVA:
- A. Reject H_0 : when critical value is greater than test statistic
 - B. Reject H_0 :when critical value is equal to test statistic
 - C. Reject H_0 :when critical value is either greater than or less than test statistic
147. Do not reject H_0 : when critical value is greater than test statistic Of the following one is not the features of operations research.
- A. Methodological Approach
 - B. Inter-disciplinary approach
 - C. Objectivistic Approach
 - D. None of the above.
148. In graphic method the restriction of number of variables is ;
- A. 2
 - B. 3
 - C. Not more than 3
 - D. None of the above.
149. The following define the corner points of a feasible region with decision variables labelled as x (horizontal axis) and y (vertical axis). $([x, y]: 0,0; 0,15; 25,0)$. which of the listed constraints is included within the feasible region?
- A. $3x+5y < 75$
 - B. $3x+5y > 75$
 - C. $5x+3y < 75$
 - D. $5x+3y > 75$
150. Which one is the correct equation for calculating the early finish time (EFT) for an activity in a project?
- A. $EFT = EFT - LST$
 - B. $EFT = LFT + \text{duration}$
 - C. $EFT = EST + \text{Duration}$
 - D. $EFT = LFT - ES$

151. The total number of allocation in a basic feasible solution of transportation problem of $m \times n$ size is equal to:
- A. $m \times n$
 - B. $(m / n) - 1$
 - C. $m + n + 1$
 - D. $m + n - 1$
152. An activity that has probabilistic completion time of 20, 30, 46. What is its mean time?
- A. 21
 - B. 31
 - C. 16
 - D. 43
153. If the primal has unbounded solution , then the dual has
- A. Optimal solution
 - B. No solution
 - C. Bounded solution
 - D. None of the above
154. One of the following is true about the assumption of assignment model
- A. Number of jobs is equal to the number of machines or persons
 - B. Each man or machine is loaded with more than one job at a time.
 - C. Each man or machine is independently capable of handling any of the job being presented
 - D. All of the above.
155. Are a set of mutually exclusive and collectively exhaustive decisions that are available the decision maker.
- A. Decision criteria
 - B. Pay-off matrix
 - C. Alternatives
 - D. State of nature
156. In game theory, a situation in which one firm can gain only what another firm loses is called
- A. Nonzero-sum game.
 - B. Prisoners' dilemma.
 - C. Zero-sum game.
 - D. Cartel temptation